



Residential Smart Grid

Report - 02

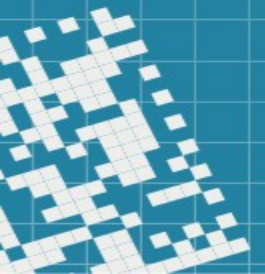
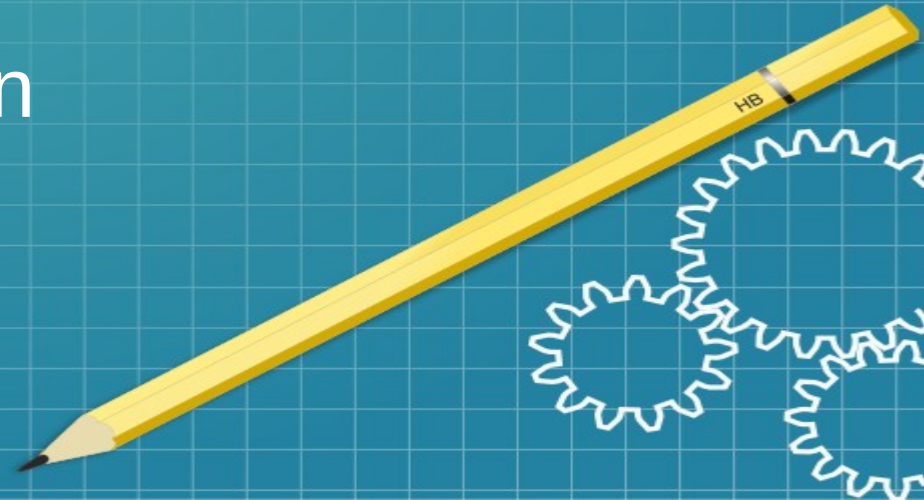
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Project Recap

- Project Description
- System Design
- Prototype System Design

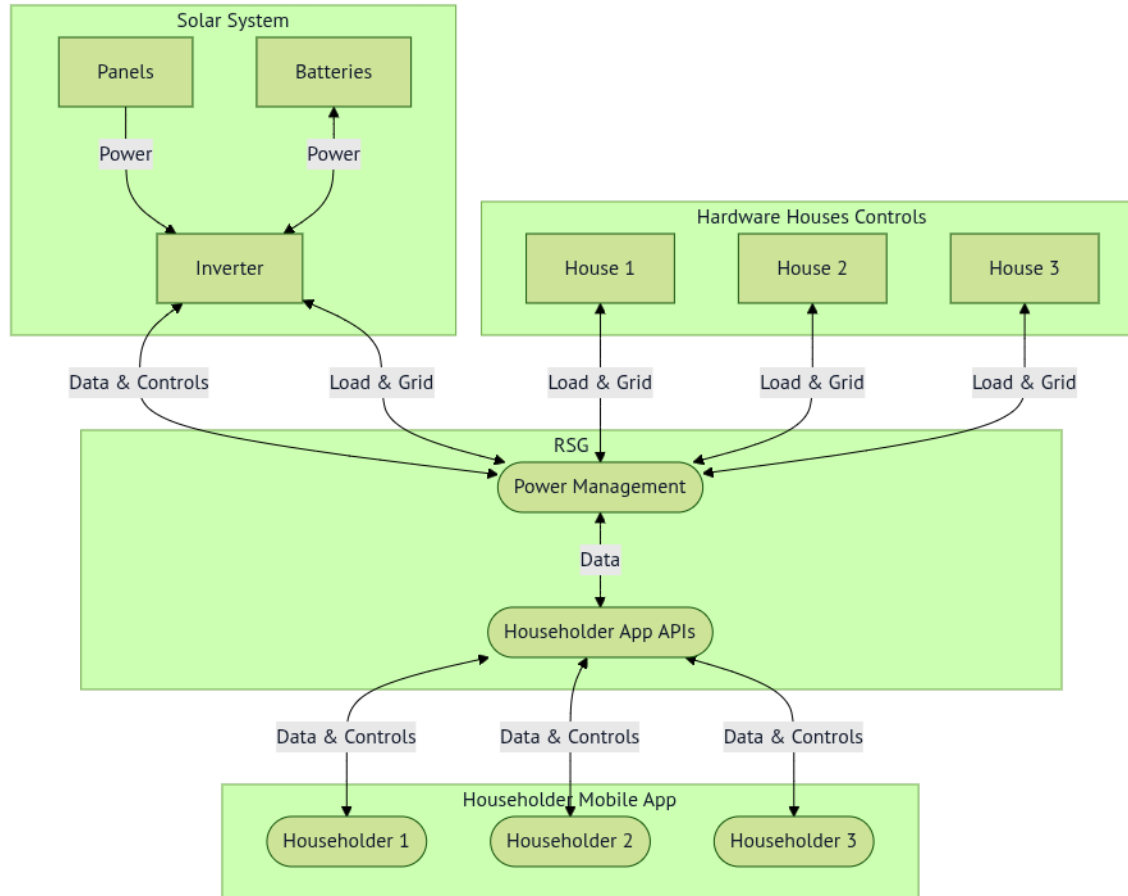


Project Description

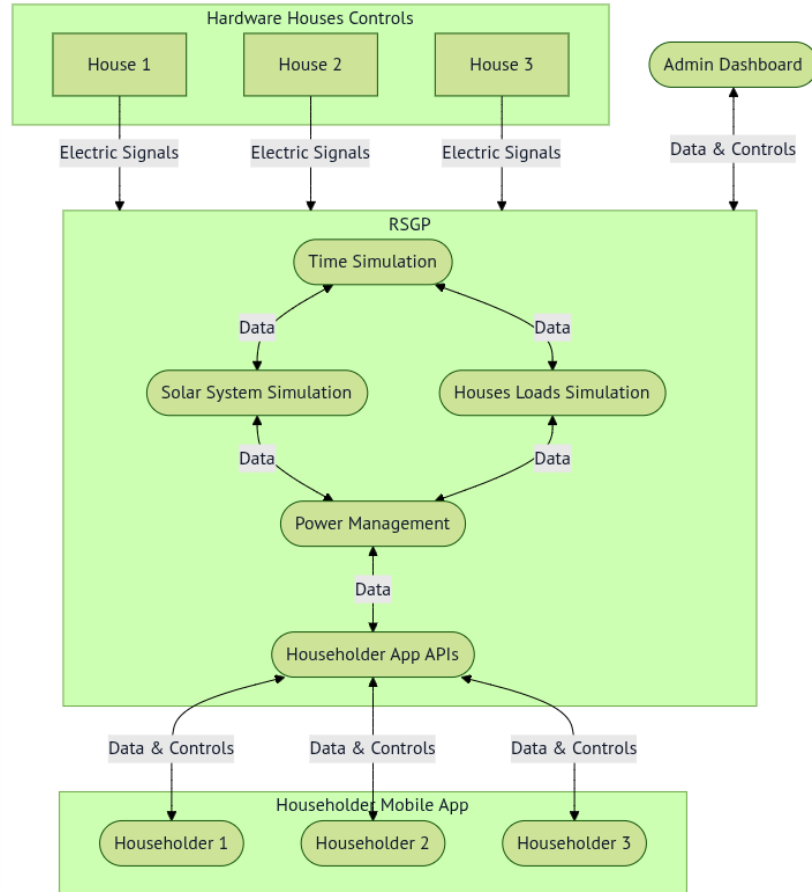


- RSG (Residential Smart Grid) is a grid system that links a large PV system for a residential building with each house in that building. This system controls the power flow between the PV system and each house load and grid lines, using smart techniques and methods. Its aim is to minimize grid dependency and excess PV power.
- Along that grid system is an admin dashboard and a mobile app for each householder to browse and control its own system.

System Design



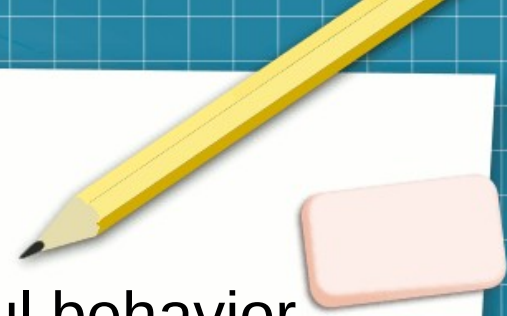
Prototype System Design



Current Progress & Plans

- RSGP
 - Time Simulation
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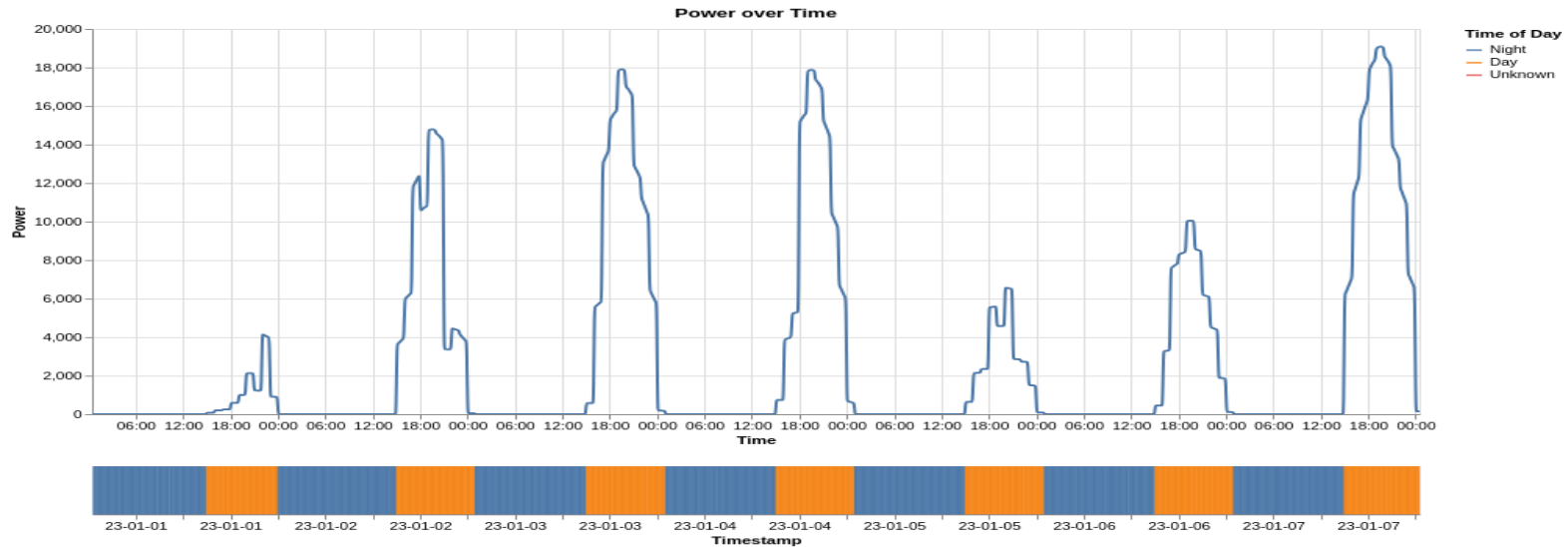
RSGP - Time Simulation



- Simulates time going faster to allow for meaningful behavior that takes hours to appear, and syncs that time between the other parts in the system.
- Removed Location from the simulation as it is now fixed to be Arizona, due to restrictions on where solar data is collected. We have chosen Arizona because its climate is similar to ours.
- No future plans exist for this.

RSGP - Solar System Simulation

- Uses real data from NSRDB for Arizona. The data includes DNI, DHI, GHI, Zenith Angle, and other related information that is used to calculate the power generated from the panels, using pvlib in Python.



RSGP – Houses Loads Simulation

- No changes to the software simulation; no progress on the hardware simulation yet.

RSGP - Power Management

- Only charges and discharges the battery at this point.
- Power Management Solutions were planned to be implemented incrementally:
 1. Static and equal virtual battery.
 2. Dynamic and initially equal virtual battery; adjusted through usage trends through a mathematical model.
 3. Dynamic and initially equal virtual battery; adjusted through AI House Load prediction and weather forecasting.
 4. Dynamic and initially equal virtual battery with an energy pool; adjusted through AI House Load prediction and weather forecasting.
 5. Reinforcement Learning.

RSGP - Householder App API

- Not implemented yet; currently under consideration to be implemented along a mobile app.

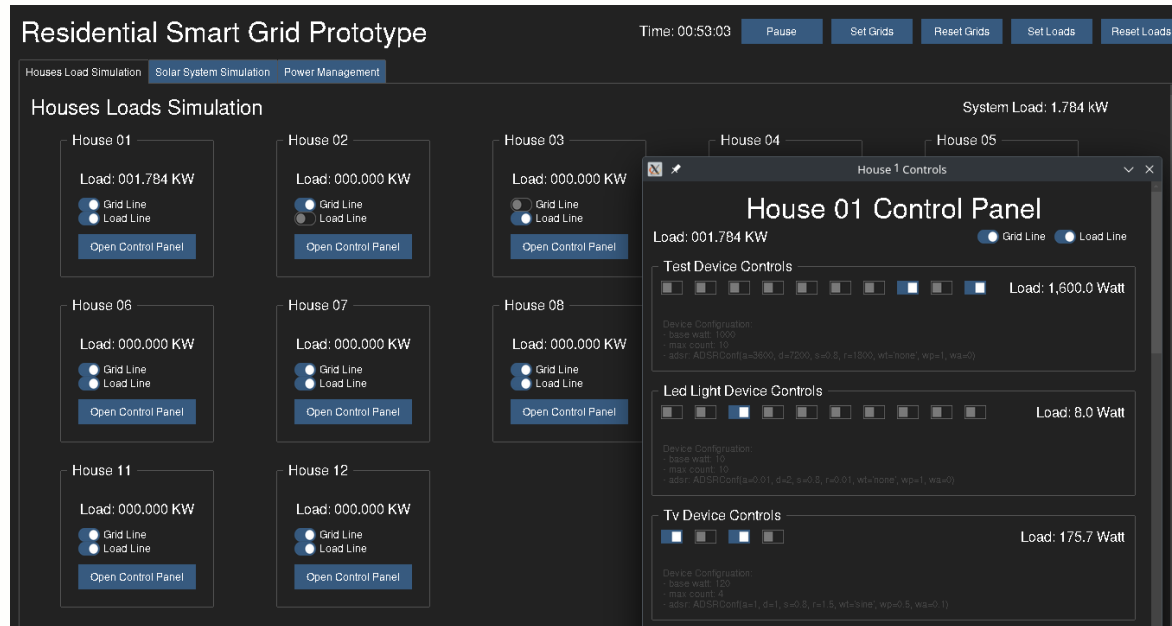
Householder Mobile App

- Some initial work and designs have been implemented, but nothing yet to be called progress.



Admin Dashboard

- No changes. Currently under maintenance to be split from the RSGP process, to be ran on a separate machine.

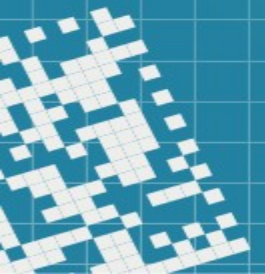
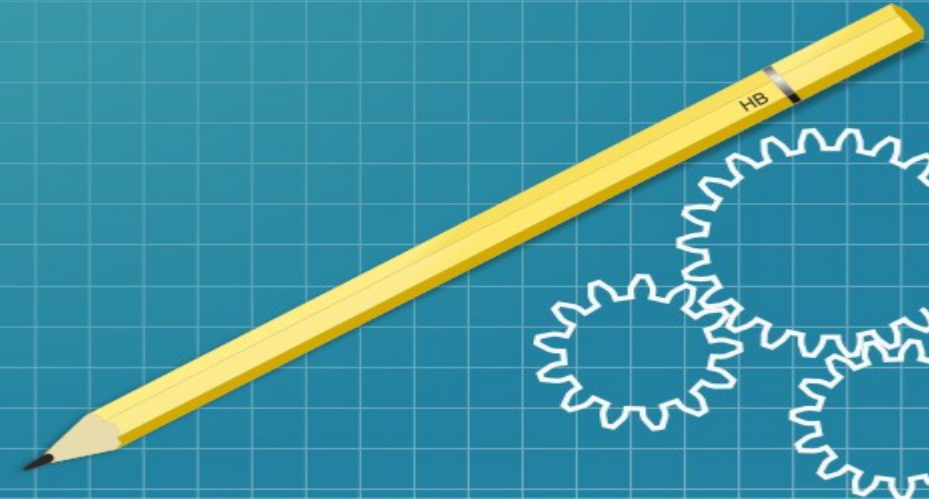


Notebooks

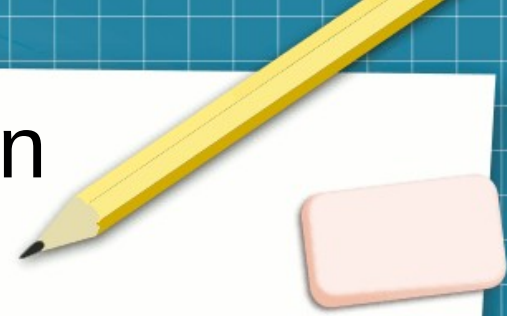


- Revamped the notebooks and added visualization for the NSRDB data.
- Currently, there exists four notebooks:
 - Graphs Editor
 - Inverter Logs Visualization
 - RSGP Logs Visualization
 - NSRDB Visualization

Documentation & Dissertation



Documentation & Dissertation



- Changed the outline to the following:
 - Chapter 01: Introduction
 - Chapter 02: Project Preview
 - Chapter 03: Time Simulation
 - Chapter 04: Solar System Simulation
 - Chapter 05: Houses Loads Simulation
 - Chapter 06: Power Management
 - Chapter 07: Householder App API
 - Chapter 08: Householder Mobile App
 - Chapter 09: Final Results
 - Chapter 10: Conclusions and Suggestions

Documentation & Dissertation

- Documented the currently worked part of Houses Loads Simulation; the software part of the simulation.

